

Lessons from a Gravity Campaign: Why Newtonian Medium Theories Are Easy and Einsteinian Ones Are Hard

A framework-independent distillation of an exhaustive audit of classical gravitational mechanisms in an elastic-network model. July 2026.

Abstract

We report general lessons from a systematic campaign to derive weak-field gravity inside a mechanical medium theory (an elastic network of tensioned filaments). The campaign failed at its stated objective and succeeded at a different one: it produced a small set of theorems explaining why medium theories reproduce Newtonian gravity almost for free, why they generically fail at Einstein's weak-field metric, and exactly what any emergent-gravity proposal — in any framework — must accomplish to succeed where this one did not. The results are stated here without reference to the parent framework's particulars, because we believe they apply to the entire class of theories in which gravity is an effective property of a material substrate.

1. Why Newtonian gravity is nearly free

Any medium that attaches conserved stress to masses delivers an inverse-square force by geometry alone. If mass-defects anchor tension, absorb an ambient flux, or source a static conserved field, then the total of that quantity crossing any sphere around the mass is fixed, and its surface density falls as $1/(4\pi r^2)$. Force balance of the medium (the divergence of stress equaling the applied force density) is the Poisson equation in disguise, so the $1/r$ potential and $1/r^2$ force follow from statics, and no topology of branching, merging, or fanning escapes the conservation. Action-at-a-distance dissolves because the connection pre-exists. This is why the history of physics is littered with qualitatively successful mechanical accounts of Newtonian attraction — Le Sage shadows, tension networks, flux screens: Newton's phenomenology is the fixed point of conservation plus three-dimensional geometry, and nearly any substrate lands on it.

2. What Einstein's weak field actually demands

The measured weak field is not a force law but a metric structure, and its content can be reduced to two operational statements: clocks slow near mass in proportion to a potential falling as $1/r$, and the effective speed of light is modulated anisotropically — twice as strongly along the radial direction as tangentially — at that same $1/r$ order. Decomposed, the required conditioning of the medium is an isotropic part plus a traceless (quadrupolar) part locked at exactly one quarter of the isotropic amplitude, both of potential range, at all radii. The factor of two in light deflection, the full Shapiro delay, and the PPN parameter $\gamma = 1$ (verified to a few parts in 10^5) are one and the same demand. A viable mechanism must therefore transport directional information about the source out to arbitrary distance at potential order, with a ratio it cannot adjust.

3. The obstruction: metric order versus strain order

Here is the theorem that killed the classical campaign, in framework-independent form. In a material substrate, local properties can respond only to strain — gradients of the displacement field — because uniform displacement is unobservable (a gauge freedom every substrate shares). Outside a localized source, the static conditioning potential is harmonic, so its slowest-decaying displacement falls as $1/r^2$ and the strain it produces falls as $1/r^3$: TIDAL order. The metric lives one derivative lower, at POTENTIAL order, $1/r$. The substrate thus reproduces the correct tensor SHAPE exactly — the Hessian of $1/r$ has eigenvalues proportional to $(2, -1, -1)$, which is precisely the traceless structure the Einstein weak field contains — but one derivative too high. The medium possesses general relativity's tidal geometry natively and cannot promote it to metric range by any elastic means.

4. The solid-angle dilution theorem

The only structures that reach potential range are direction-blind. A scalar field that diffuses through the medium (an ambient-flux deficit, a temperature-like agitation) obeys Laplace's equation and falls as $1/r$ — but diffusion erases directional memory, so it conditions isotropically. Conversely, every direction-carrying structure near a mass — the population of filaments, shadows, or stress lines anchored to it — dilutes as $1/r^2$, because its local fraction is a solid-angle fraction. Ballistic transport keeps direction and loses range; diffusive transport keeps range and loses direction. The two regimes bracket Einstein's required quadrupole-to-isotropic ratio (which lies strictly between them) without either being able to hold it, and holding it at all radii would require scale-free transport, which ordinary substrates do not provide.

5. The generic scalar-medium prediction: $\gamma \approx -1/2$

A corollary explains a long historical pattern. A medium conditioned isotropically in its wave-carrying stiffness speeds light up (or slows it) equally in all directions while shifting clocks and rulers in a fixed conspiracy; working out the effective PPN parameter gives $\gamma = -1/2$ for pure stiffness-conditioning, 0 for pure density, and -1 for pure rigidity — every isotropic channel lies in $[-1, 0]$. This is why mechanical and emitter theories of gravity have always managed orbits and redshift and always failed light deflection: they were scalar mimics, structurally confined to the wrong side of $\gamma = 0$, predicting roughly a quarter of the observed bending. The 1919 eclipse did not merely disfavor such theories; it excluded a class.

6. What any emergent-gravity proposal must overcome

Collecting the results as requirements: (i) produce conditioning at potential order ($1/r$), not tidal order; (ii) transport directional information to that range despite solid-angle dilution; (iii) lock the traceless-to-isotropic ratio at one quarter, scale-independently, without a tunable parameter; and (iv) do all of this while leaving the substrate's uniform state gravitationally invisible, or its rest energy will curve everything catastrophically. Requirement (iv) is generically satisfiable (metric perturbations are functionals of deviations); requirements (i)–(iii) jointly exclude, so far as we could determine, every classical mechanism constructible in an elastic substrate. The standard escape is Sakharov induction — letting quantum fluctuations of the substrate's modes generate the metric's covariant dynamics at loop level — whose known

difficulty is that generic media induce non-covariant dynamics; a substrate whose matter-coupled sector is exactly Lorentz invariant is the natural candidate to evade that, and computing the induced action for such a substrate is, in our view, the sharpest open problem this campaign leaves behind.

7. A methodological remark

These lessons exist because the campaign was permitted to fail. Each candidate mechanism was computed to its exact prediction, compared to measurement, and retained as a documented negative when it lost; a postulate that would have rescued the conclusion was declined because its only motivation was the rescue. We suggest that the value of speculative frameworks is best measured not by the phenomena they claim, but by whether their failures are sharp enough to become theorems. Under that measure, a falsified gravity sector produced this document — which we believe says something general, and useful, to anyone attempting to build spacetime from a substrate.

A mechanism's Newtonian success and its Einsteinian failure can be the same mathematical fact.